

Lisa Frueh, MPH

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Education

- PhD Drexel Dornsife School of Public Health** Philadelphia, PA
Department: Environmental and Occupational Health Expected 2026
Mentor: Jane E. Clougherty, MsC, ScD
- MPH Harvard T.H. Chan School of Public Health** Boston, MA
Field of Study: Social and Behavioral Sciences Sept. 2019 - March 2021
Concentration: Women, Gender, and Health
- BA Reed College** Portland, OR
Major: Chemistry Sept. 2011 - May 2015
Thesis: A contribution to the fight against malaria: novel synthesis and characterization of 3-alkylarylether-4(1*H*)-quinolones with antimalarial properties

Research & Training Interests

I am interested in understanding how the institutions we interact with (like schools and workplaces) shape our susceptibility and resilience to ambient environmental exposures. To explore this topic, I leverage methodological expertise in exposure assessment and environmental epidemiology and a commitment to environmental justice. I value community engagement and science communications in my work and seek future opportunities to continue such engagement.

Research Experience

Drexel University Dornsife School of Public Health, Philadelphia, PA

National Research Service Award Predoctoral Fellow Aug. 2025 – Present
National Institute of Environmental Health Sciences

Designed, and am currently completing, an NIH-funded dissertation project exploring how the school environment modifies pediatric susceptibility to PM_{2.5} and heat exposure in New York State. Analyzing publicly available school quality data and linkages to electronic health records and environmental exposures estimates to explore these associations.

Research Assistant Sep. 2025 – Present
Dept. of Epidemiology | Dr. Jacquelyn Jahn

Leading a research project investigating the link between temperature and mortality in California state prisons.

Doctoral Research Fellow Sep. 2022 – Aug. 2025
Dept. of Environmental and Occupational Health | Dr. Jane Clougherty

Supported research activities related to two projects: 1) developing and implementing a community-based air monitoring protocol in a South Philadelphia fenceline community, including developing targeted report-backs and public web apps to share results; 2) identifying and analyzing GIS-based social stressors and asset indicators in upstate New York and assigning air pollution and temperature

exposures to a large, geocoded health dataset. Primary responsibilities included managing and cleaning data and performing epidemiologic analyses investigating the link between climate-related stressors and pediatric health and the role of resilience factors in modifying the relationship.

Inclusion, Diversity, Equity, & Antiracism (IDEA) Fellow

May 2023 – Sep. 2023

Urban Health Collaborative & Dept. of Epidemiology | Dr. Gina Lovasi

Developed online GIS-based resources for non-academics to visually explore select metrics of spatial social polarization. Contributed to the authorship of a systematic review on spatial social polarization measures. Available at: https://drexel-uhc.shinyapps.io/SSP_Maps/

Harvard T.H. Chan School of Public Health, Boston, MA

Research Assistant

Feb. 2021 – Sept. 2022

Dept. of Environmental Health | Dr. Francine Laden

Developed online GIS-based resources for community members about environmental racism in greater Boston across four domains: residential segregation, neighborhood exposures, household exposures, and personal care products.

Brigham and Women's Hospital, Boston, MA

Research Assistant

Nov. 2019 – May 2022

Channing Division of Network Medicine | Dr. Heather Eliassen

Wrote literature reviews, reviewed manuscripts, and supported research activities in the Nurses' Health Studies, which investigate diet, lifestyle, and social risk factors associated with major chronic diseases in women. Administrative duties included website maintenance (external and internal) and co-coordinating the annual newsletter.

Project Weber / RENEW (PWR) & Brown University, Providence, RI

Research Team Volunteer

May 2021 – May 2022

Brown University Dept. of Epidemiology | Dr. Alexandria Macmadu

Extracted qualitative and quantitative data from police reports of overdose calls in 2019/2020 in Providence, RI. Worked with a team of researchers at Brown University School of Public Health and advocates from harm-reduction organization PWR to develop an evidence base for future policy recommendations to limit police response to overdose calls in favor of Fire/Rescue. Published findings in academic papers as well as policy briefs.

Massachusetts Department of Public Health, Boston, MA

Summer Intern

May 2020 – Sept. 2020

Bureau of Community Health and Prevention

Authored a report summarizing results of COVID-19 impact surveys from around the country with a health equity focus, including a repository of ongoing surveys and recommendations for equity-focused pandemic response. Presented findings to stakeholders at the MA Dept. of Public Health.

Oregon Health and Science University, Portland, OR

Research Assistant

Sep. 2018 – Aug. 2019

Dept. of Emergency Medicine | Dr. Brandon Maughan

Built and managed online survey data collection and participant management database in REDCap for a study on quality of life after pulmonary embolism, collaborated with primary investigator on IRB applications and revisions, conducted qualitative interviews, and co-authored a systematic review of relevant literature.

Research Assistant II

May 2014 – Aug. 2017

Dept. of Molecular Microbiology and Immunology | Dr. Michael Riscoe

Developed novel antimalarial drugs in a diverse team of chemists and biologists working toward a low-cost, single-dose cure for malaria, mentored summer interns and one undergraduate thesis.

Student Research Assistant

May 2012 – Sep. 2012

Dept. of Molecular Microbiology and Immunology | Dr. Mary Stenzel-Poore

Supported research activities related to inflammatory response pathways following ischemic injury.

Reed College, Portland, OR

Student Research Assistant

May 2013 – Sep. 2013

Dept. of Chemistry | Dr. Aaron Nilsen

Designed and synthesized six analogs of a promising anti-Dengue virus compound in a team of four undergraduate students, communicated results to chemistry department faculty in a written report and oral presentation at the end of the summer.

Teaching Experience

Drexel University, Philadelphia, PA

Teaching Assistantships

Exposure Science | Dr. Jane Clougherty

Spring 2024, Spring 2025

Assisted with syllabus design, assessment design, grading, and lecture instruction of Exposure Science, a graduate-level course.

Principles and Practice of Environmental and Occupational Health | Dr. Jerry Fagliano Fall 2023

Assisted with syllabus design, grading, and instruction of Introduction to Environmental and Occupational Health, a masters-level course.

Guest Lectures

Managing and Analyzing Environmental Epidemiology Data

May 26, 2025

“Designing a Useful Table One”

Principles and Practice of Environmental and Occupational Health

Oct. 3, 2024 & Oct. 2, 2025

“Exposure Distribution and Disparities: Environmental Justice”

Reed College, Portland, OR

Teaching Assistantships

Introductory Chemistry Laboratory

Fall 2013, Fall 2014

Analytical Chemistry Laboratory

Spring 2014

Tutoring

Tutor / Head Tutor

Fall 2012 – Fall 2014

Instructed on an individual basis and in groups at the tutoring center; subjects taught included eight courses in chemistry, biology, and mathematics. Additional head tutor duties involved pedagogic trainings and supervision at the tutoring center.

Grants

Current Research

Predocutorial Individual National Research Service Award

PI: Lisa Frueh

Sponsor: Jane Clougherty

Funding Organization: National Institute of Environmental Health Sciences

Amount Awarded: \$43,138

Period of Grant Award: 08/01/2025 – Present

Title: Schools as health-related assets: The role of school physical, educational, and social environments in shaping children's resilience to temperature and fine particulate matter exposure in New York

Completed Research

Peace Engineering Fellowship

PI: Lisa Frueh

Funding Organization: Build Up

Amount Awarded: \$6,000

Period of Grant Award: 06/01/2023 – 06/01/2024

Title: Evaluating report-back of results from an air monitoring network in industrialized South Philadelphia communities

Awards

Drexel University Dornsife School of Public Health Urban Health Collaborative

2025

Dissertation Award

Awarded to two Dornsife doctoral candidates annually, the Urban Health Collaborative (UHC) supports additional training or research funding up to \$5,000 per awardee.

Mid-Atlantic Regional Public Health Training Center

2024

Faculty-Student Collaborative Project

Administered through the University of Pittsburgh, Collaborative Projects provide funding for students to work with faculty in practice-based research and evidence-based practice. This award provided funding for a summer-long project to work on data analysis and report-back materials for THRIVEair, an NIH-funded community-based participatory air monitoring study in South and Southwest Philadelphia.

International Society of Environmental Epidemiology

2024

Wartenberg Travel Award

Awarded to 16 researchers engaged in community-engaged environmental epidemiology. Funded travel, accommodations, and registration to the 2024 International Society of Environmental Epidemiology (ISEE) annual conference in Santiago, Chile.

Drexel University

Graduate College Doctoral Fellowship – Gold Level

2022 – 2024

University-wide, faculty-nominated fellowship for graduate students. Gold level funding (\$10,000 award) is awarded to 2-3 students annually.

Drexel University Dornsife School of Public Health*Dean's Doctoral Fellowship*

2022 – 2024

School-wide, faculty-nominated fellowship that provided full stipend support for the first two years of my doctoral training. I was the only recipient of this award in 2022 and was re-appointed for the 2023-2024 academic year.

Drexel University Dornsife School of Public Health*Inclusion, Diversity, Equity & Antiracism (IDEA) Fellowship*

2023

School-wide fellowship that funded summer work on a project to create interactive, GIS-based resources about measures of spatial social polarization.

Harvard University*MPH Scholarship*

2019 – 2021

Merit-based scholarship covering 50% of tuition and fees.

Honors

Interdisciplinary Association for Population Health Sciences*Best Poster Award*

2025

One of 6 winners awarded at the IAPHS 2025 conference.

Drexel Dornsife School of Public Health*Outstanding Promise Student Award*

2024

School-wide award that recognizes one student in each academic level (undergraduate, master's, and doctoral) who has "exhibited great promise for regional or national societal impact, and accordingly, for enhancing Drexel's reputation, now and in the future." Based on faculty and student nomination.

Drexel University*Teaching Assistance Excellence Award*

2024

Based on faculty and student nomination, awarded to 10 teaching assistants who exhibit exemplary commitment to student learning through reflective teaching practices, creative and innovative teaching methods, academic support, leadership and a commitment to their own professional growth and development as an educator.

Graduate College Recognition Award

2024

Awarded to leaders of the Public Health Doctoral Students Advisory Group (PHDSAG), for leadership and advocacy in support of the graduate student community's health and well-being and efforts to enhance the student experience.

ESRI*Runner-Up, ArcGIS Online StoryMap Competition*

2022

The 2022 ArcGIS StoryMaps Competition "invited storytellers globally to build stories about an idea, challenge, or solution through innovative data visualization techniques using ArcGIS StoryMaps." There was one winner and one runner-up in the Humanitarian and Social Justice category.

Reed College*Phi Beta Kappa*

2015

National honors society, with membership based on recommendation from participating school.

Faculty-nominated commendation for academic achievement, awarded annually.

Service

Drexel University Dornsife School of Public Health

Public Health Doctoral Students Advisory Group (PHDSAG) Executive Board

President

June 2024 – Present

Facilitating biweekly board meetings, quarterly meetings with administration, and regular organization events.

Event Coordinator

June 2023 – June 2024

Organized events for doctoral students in the school, coordinating with other board members, submitting event documentation and budget requests. Attending biweekly board meetings and events.

Department of Environmental and Occupational Health

Journal Club Committee Co-Chair

April 2023 – Present

Organizing monthly journal clubs in Environmental and Occupational Health.

Seminar Committee Co-Organizer

Sept. 2022 – Present

Selecting speakers for seminars departmental seminars in Environmental and Occupational Health with two faculty members.

Professional Activities

Editorial Duties

Journal of Exposure Science & Environmental Epidemiology (JESEE) – Reviewer	2025
Inquiry – Reviewer	2025
Social Science and Medicine – Reviewer	2024
Environmental Justice Journal – Reviewer	2023
Science of the Total Environment (STOTEN) – Reviewer	2022
American Public Health Association – Abstract reviewer	2021

Media Exposure

1. Rowhomes versus refineries: Academics and community members are working together to tackle environmental hazards as part of Drexel's new Environmental Collaboratory, a pioneering initiative that funds research activism. *Drexel Magazine*. 2023. Available at: <https://drexelmagazine.org/2023/rowhomes-versus-refineries/>
2. Show and tell: Interactive map tells story of environmental racism in Boston. *HSPH Center for Health Communications Blog*. 2023. Available at: <https://www.hsph.harvard.edu/chc/2023/03/02/show-and-tell-interactive-map-tells-story-of-environmental-racism-in-boston/>
3. Project on racial residential segregation runner-up in data viz competition. *HSPH News*. 2023. Available at: <https://www.hsph.harvard.edu/news/hsph-in-the-news/project-on-racial-residential-segregation-runner-up-in-data-viz-competition/>
4. University research examines role of police at overdose incidents. *Brown Daily Herald*. 2022. Available at: <https://www.browndailyherald.com/article/2022/11/university-research-examines-role-of-police-at-overdose-incidents>

5. Get inspired by the 2022 ArcGIS StoryMaps Competition winners—a showcase of stunning data viz. *Esri ArcGIS Blog*. 2022 Available at: <https://www.esri.com/arcgis-blog/products/arcgis-storymaps/constituent-engagement/get-inspired-by-the-2022-arcgis-storymaps-competition-winners/>
6. At overdose events, arrests by police and combative behavior are rare, study finds. *Brown University News*. 2022. Available at: <https://www.brown.edu/news/2022-11-07/police-overdose>
7. Interactive series explores environmental racism. *HSPH News*. 2022. Available at: <https://www.hsph.harvard.edu/news/features/interactive-web-series-explores-environmental-racism/>

Professional Development

Columbia University, New York, NY Aug. 14-15, 2025
Skills for Health and Research Professionals (SHARP) Training I Bayesian Modeling for Environmental Health

Drexel University, Philadelphia, PA Sep. 2024 – June 2025
Non-Credit Certificate in College Teaching

Columbia University, New York, NY July 1-2, 2024
Skills for Health and Research Professionals (SHARP) Training I Code Rigor and Reproducibility in R

Columbia University, New York, NY (virtual) Aug. 2-3, 2021
Skills for Health and Research Professionals (SHARP) Training I Environmental Justice Boot Camp: Theory and Methods to Study Environmental Health Disparities

Skills

Research: Literature review, project design, data science including regression analysis, spatial statistical methods, survey design, participant recruitment, qualitative interviewing

Computer: R, Python, ArcGIS, R Shiny, ArcGIS Online, HTML/CSS

Languages: English (first language), German (intermediate), Spanish (novice)

Peer-Reviewed Publications

1. Tripathy S, **Frueh L**, Moore K, Nguyen JA, Sanders S, Aikens SL, White C, Foy CA, Robinson DM, Clincy M, Heath C, Mullison J, Winslow P, Johnson C, Tiegs G, Wahl KC, Johnston L, Johnston N, Clougherty JE. THRIVEair: A community-based air monitoring network design in a pollution-burdened Philadelphia neighborhood to advance environmental justice. [Accepted manuscript | *Environmental Justice*]
2. **Frueh L**, Sharma S, Kinnee EJ, Kubzanksy LD, Sheffield PE, Clougherty JE. Community assets as protective factors buffering health effects of environmental hazards: framework and considerations for environmental epidemiologists. *Environmental Research: Health*. 2026;4:013002. doi: 10.1088/2752-5309/ae3c3c
3. McCulley EM, **Frueh L**, Myers D, Jaros D, Abdel Magid HS, Bayer F, Lovasi GS. Measuring Spatial Social Polarization in public health research: A scoping review. *Journal of Urban Health*. 2025;102:213-239. doi: 10.1007/s11524-024-00957-6

4. **Frueh L**, Sharma R, Sheffield PE, Clougherty JE. Community violence and asthma: A review. *Annals of Asthma, Allergy & Immunology* 2024;133(6):641-648.e12. doi: 10.1016/j.anai.2024.07.016
5. Sharma R, Humphrey JL, **Frueh L**, Kinnee EJ, Sheffield PE, Clougherty JE. Neighborhood violence and socioeconomic deprivation influence associations between acute air pollution and temperature on childhood asthma in New York City. *Environmental Research* 2023;213(3):116235. doi:10.1016/j.envres.2023.116235
6. Zeleznik OA, Kang JH, Su J, **Frueh L**, Elze T, Khawaja A, Wiggs JL, Clish C, Rosner B, Pasquale L, Eliassen H. Plasma Metabolite Profile for Primary Open-angle Glaucoma in Three US Cohorts and the UK Biobank. *Nature Communications* 2023;14(1):2860. doi: 10.1038/s41467-023-38466-w
7. **Frueh L**, Collins AB, Newman R, Barnett NP, Rich JD, Clark MA, Marshall BDL, Macmadu A. Multi-level influences on increased overdose risk behaviors during the COVID-19 pandemic among people who use drugs in Rhode Island: a qualitative investigation. *Harm Reduction Journal*. 2023;20(1):14. doi:10.1186/s12954-023-00741-w
8. Zeleznik OA, Welling DB, Stankovic K, **Frueh L**, Balasubramaniam R, Curhan GC, Curhan SG. Association of Plasma Metabolomic Biomarkers With Persistent Tinnitus: A Population-Based Case-Control Study. *JAMA Otolaryngology–Head & Neck Surgery* 2023;149(5):404-415. doi:10.1001/jamaoto.2023.0052
9. Wang T, Heng YJ, Baker GM, Bret-Mounet VC, Quintana LM, **Frueh L**, Hankinson SU, Holmes MD, Chen WY, Willet WC, Rosner B, Tamimi RM, Eliassen HA. Loss of PTEN Expression, PIK3CA Mutations, and Breast Cancer Survival in the Nurses' Health Studies. *Cancer Epidemiol Biomarkers Prev*. 2022;31(10):1926-1934. doi:10.1158/1055-9965.EPI-22-0672
10. Macmadu A, Yolken A, **Frueh L**, Touissant JR, Newman R, Jacka BP, Collins AB, Marshall BDL. Characteristics of events in which police responded to overdoses: an examination of incident reports in Rhode Island. *Harm Reduct J*. 2022;19(1):1-10. doi:10.1186/s12954-022-00698-2
11. Kang JH, Zeleznik O, **Frueh L**, Lasky-Su J, Eliassen HA, Clish C, Rosner BA, Pasquale LR, Wiggs JL. Prediagnostic Plasma Metabolomics and the Risk of Exfoliation Glaucoma. *Investigative Ophthalmology & Visual Science*. 2022;63(9):15. doi:10.1167/iov.63.9.15
12. Kang JH, Wang M, **Frueh L**, Rosner B, Wiggs JL, Elze T, Pasquale LR. Cohort Study of Race/Ethnicity and Incident Primary Open-Angle Glaucoma Characterized by Autonomously Determined Visual Field Loss Patterns. *Translational Vision Science & Technology*. 2022;11(7):21. doi:10.1167/tvst.11.7.21
13. Macmadu A, **Frueh L**, Collins AB, Newman R, Barnett NP, Rich JD, Clark MA, Marshall BDL. Drug use behaviors, trauma, and emotional affect following the overdose of a social network member: A qualitative investigation. *Int J Drug Policy*. 2022;107:103792. doi:10.1016/j.drugpo.2022.103792
14. Pal AC, Renard I, Singh P, Vydyam P, Chiu JE, Pou S, Winter RW, Dodean R, **Frueh L**, Nilsen AC, Riscoe MK, Doggett S, Mamoun BC. *Babesia duncani* as a model organism to study the development, virulence and drug susceptibility of intraerythrocytic parasites *in vitro* and *in vivo*. *Journal of Infectious Diseases*. Published online May 4, 2022;jiac181. doi:10.1093/infdis/jiac181

15. Mínguez-Alcarón L, **Frueh L**, Williams PL, James-Todd T, Souter S, Ford JB, Rexrode KM, Calafat AM, Hauser R, Chavarro JE, The Earth Study Team. Pregnancy urinary concentrations of bisphenol A, parabens and other phenols in relation to serum levels of lipid biomarkers: Results from the EARTH study. *Science of the Total Environment*. 2022;833(10):155191. doi: 10.1016/j.scitotenv.2022.155191
16. Pou S, Dodean RA, **Frueh L**, Liebman KM, Gallagher RT, Jin H, Jacobs RY, Nilsen A, Stuart DR, Doggett JS, Riscoe MK, Winter RW. A New Scalable Synthesis of ELQ-300, ELQ-316, and other Antiparasitic Quinolones. *Organic Process Research & Development*. 2021;25(8):1841-1852. doi:10.1021/acs.oprd.1c00099
17. Chiu JE, Renard I, Pal AC, Singh P, Vydyam P, Thekkiniath J, Kumar M, Gihaz S, Pou S, Winter RW, Dodean R, **Frueh L**, Nilsen AC, Riscoe MK, Doggett JS, Mamoun CB. Effective Therapy Targeting Cytochrome bc1 Prevents Babesia Erythrocytic Development and Protects from Lethal Infection. *Antimicrobial Agents and Chemotherapy*. 2021;65(9):e0066221. doi:10.1128/AAC.00662-21
18. Zeleznik OA, Balasubramanian R, Zhao Y, **Frueh L**, Jeanfavre S, Avila-Pachecho J, Clish CB, Tworoger S, Eliassen HA. Circulating amino acids and amino acid-related metabolites and risk of breast cancer among predominantly premenopausal women. *NPJ Breast Cancer*. 2021;7(1):1-10. doi:10.1038/s41523-021-00262-4
19. Zeleznik OA, Balasubramanian R, Ren Y, Tobias DK, Rosner BA, Peng C, Bever AM, **Frueh L**, Jeanfavre S, Avila-Pachecho J, Clish CB, Mora S, Hu FB, Eliassen HA. Branched-Chain Amino Acids and Risk of Breast Cancer. *JNCI Cancer Spectrum*. 2021;5(5):pkab059. doi:10.1093/jncics/pkab059
20. Maughan BC, **Frueh L**, McDonagh MS, Casciere B, Kline JA. Outpatient Treatment of Low-risk Pulmonary Embolism in the Era of Direct Oral Anticoagulants: A Systematic Review. *Academic Emergency Medicine*. 2021;28(2):226-239. doi: 10.1111/acem.14108
21. Potharaju S, Mutyam SK, Liu M, Green C, **Frueh L**, Nilsen A, Pou S, Winter R, Riscoe MK, Shankar G. Improving solubility and oral bioavailability of a novel antimalarial prodrug: comparing spray-dried dispersions with self-emulsifying drug delivery systems. *Pharmaceutical Development and Technology*. 2020;25(5):625-639. doi:10.1080/10837450.2020.1725893
22. Smilkstein MJ, Pou S, Krollenbrock A, Bleyle LA, Dodean RA, **Frueh L**, Hinrichs DJ, Li Y, Martinson T, Munar MY, Winter RW, Bruzal I, Whiteside S, Nilsen A, Koop DR, Kelly JX, Kappe SH, Wilter BK, Riscoe MK. ELQ-331 as a prototype for extremely durable chemoprotection against malaria. *Malaria Journal*. 2019;18(1):291. doi:10.1186/s12936-019-2921-9
23. Smith JL, Sheridan K, Parkins CJ, **Frueh L**, Jemison AL, Strode K, Dow G, Nilsen A, Hirsch AJ. Characterization and structure-activity relationship analysis of a class of antiviral compounds that directly bind dengue virus capsid protein and are incorporated into virions. *Antiviral Research*. 2018;155:12-19. doi:10.1016/j.antiviral.2018.04.019
24. **Frueh L**, Li Y, Mather MW, Li Q, Pou S, Nilsen A, Winter TW, Forquer IP, Pershing AM, Xie LH, Smilkstein MJ, Caridha D, Koop DR, Campbell RF, Sciotti RJ, Kreishman-Deitrick M, Kelly JX, Vesely B, Vaidya AB, Riscoe MK. Alkoxy carbonate Ester Prodrugs of Preclinical Drug Candidate ELQ-300 for Prophylaxis and Treatment of Malaria. *ACS Infectious Diseases*. 2017;3(10):728-735. doi:10.1021/acsinfecdis.7b00062

Other Publications

Theses

1. **Frueh L.** A contribution to the fight against malaria: novel synthesis and characterization of 3-alkylarylether-4(1H)-quinolones with antimalarial properties. Undergraduate thesis. Reed College, May 2015.

Patents

1. Riscoe M, Nilsen A, Stickles A, Miley G, Winter R, Pou S, Li Y, Jane XK, Forquer I, Doggett S, Bruzal I, **Frueh L**, Dodean R, Smilkstein M, Alday H, Koop D, Bleyle L, inventor; Oregon Health and Science University, US Departments of Veterans Affairs, assignee. Quinolone-3-diarylethers. Patent No. WO2017015360A1; Jan 26, 2017.

Teaching Resources

1. Moller B, **Frueh L**, DePietro J, Rushovich R. (2020). Gender Analysis of COVID-19 Response. Teaching example from Women, Gender, and Health course (WGH 207), Harvard T.H. Chan School of Public Health. Available from the Harvard Case-Based Learning Initiative: <https://caseresources.hsph.harvard.edu/case-library>.
2. **Frueh L.** (2020). Sex, Gender, COVID-19, and DAGs: A 60-Minute Teaching Module for Introductory Epidemiology Lab. Teaching example from Women, Gender, and Health course (WGH 300), Harvard T.H. Chan School of Public Health. Available from the Harvard Case-Based Learning Initiative: <https://caseresources.hsph.harvard.edu/case-library>.

Conference Presentations

1. **Frueh L**, Tripathy S, Moore K, Tiegs G, Wahl KC, Johnston L, Johnston NA, Clougherty JE. Identifying Neighborhood Sources of Ambient Volatile Organic Compounds (VOCs): Results from THRIVEair, a Community-based Air Monitoring Project in South Philly. Poster presentation at: 2025 Interdisciplinary Association for Population Health Sciences (IAPHS) Conference; September, 2025; Pittsburgh, PA.
2. **Frueh L**, Kinnee EJ, Sheffield PE, Clougherty JE. The Spatial Distribution of School Environment Indicators in New York State and Co-Patterning with Sociodemographic Characteristics. Poster presentation at: 2025 Interdisciplinary Association for Population Health Sciences (IAPHS) Conference; September, 2025; Pittsburgh, PA.
3. **Frueh L**, Kinnee EJ, Just, A, Sheffield PE, Clougherty JE. Evaluating Public Schools as Health-Related Assets Buffering Adverse Health Impacts of Environmental Exposures in New York State. Poster presentation at: 2025 Drexel Urban Health Collaborative Urban Health Symposium; September, 2025; Philadelphia, PA.
4. **Frueh L**, Kinnee EJ, Just, A, Sheffield PE, Clougherty JE. The School Environment Modifies Pediatric Susceptibility to Air Pollution in New York State. Poster presentation at: 2025 International Society of Environmental Epidemiology (ISEE) / International Society of Exposure Science (ISES) Joint Conference; August, 2025; Atlanta, GA.
5. **Frueh L**, Kinnee EJ, Sheffield PE, Clougherty JE. The Spatial Distribution of School Environment Indicators in New York State and Co-Patterning with Sociodemographic Characteristics. Poster presentation at: 2025 International Society of Environmental

Epidemiology (ISEE) / International Society of Exposure Science (ISES) Joint Conference; August, 2025; Atlanta, GA.

6. **Frueh L.** THRIVEair: Reporting Back Results. Oral presentation at: 2024 International Society of Exposure Science (ISES) Conference; October, 2024; Montreal, Canada.
7. **Frueh L.** School quality, environmental hazards, and pediatric susceptibility in New York. Oral presentation at: 2024 International Society of Environmental Epidemiology (ISEE) Conference; August, 2024; Santiago, Chile.
8. Clougherty JE, **Frueh L**, Sheffield PE. Climate resilience. Panel discussion presented at: 2024 CAFÉ Climate and Health Conference; February, 2024; [virtual].
9. **Frueh L**, Sharma R, Kinnee EJ, Just AC, Sheffield PE, Clougherty JE. PM_{2.5} exposure is associated with an increased risk of pediatric ED visits in New York State, particularly at low concentrations. Poster presented at: 2023 International Society for Environmental Epidemiology (ISEE) Conference; September, 2023; Kaohsiung, Taiwan.
10. **Frueh L.** South Philadelphia Community Air Monitoring Project: Community-Based Participatory Study Design. Oral presentation at: 2023 International Society for Environmental Epidemiology North American Chapter (ISEE NAC) Conference; June, 2023; Corvallis, OR.
11. **Frueh L.** Climate change-related exposures and risk of hospitalization among children aged 0-17 in New York State, 2005-2019: Preliminary analyses from the Understanding Pediatric Susceptibility Across Temperature and Environment in New York (UPStATE NY) Study. Oral presentation at: 2023 Society for Epidemiologic Research Conference; June, 2023; Portland, OR.
12. **Frueh L**, Winter RW, Nilsen A, Kelly JX, Li Y, Smilkstein MJ, Potharaju S, Mutyam S, Shankar G, Riscoe MK. Novel prodrugs and formulations of antimalarial preclinical candidate ELQ- 300. Poster presented at: 2016 American Society of Tropical Medicine and Hygiene 65th Annual Meeting; November, 2016; Atlanta, GA.

Invited Presentations

1. **Frueh L.** Schools as community assets in New York: UHC Dissertation Award Overview. Oral presentation at: Drexel University Urban Health Collaborative Weekly Forum; March, 2025; Philadelphia, PA.
2. **Frueh L.** Communicating environmental health: from interactive web resources for broad audiences to targeted community air monitoring report-backs. Oral presentation at: OHSU-PSU School of Public Health; July 2024; Portland, OR.
3. **Frueh L.** Building an online interactive mapping resource to visualize spatial social polarization at varying scales: Challenges and opportunities. Oral presentation at: Drexel University Urban Health Collaborative GIS Day; November, 2023; Philadelphia, PA.
4. **Frueh L.** A Brief History of Redlining in Greater Boston. Oral presentation at: Metropolitan Area Planning Council; October, 2022; Boston, MA [virtual].

5. **Frueh L.** Communicating Environmental Justice Research: The NIEHS Center Environmental Racism StoryMap. Oral presentation at: Harvard T.H. Chan School of Public Health Environmental Justice Student Organization; March, 2022; Boston, MA.
6. **Frueh L.** Development of an interactive, GIS-based online resource for a broad audience. Oral presentation at: Harvard Chan NIEHS-Center SPARC Session; February, 2022; Boston, MA [virtual].